

Time Column Data Dictionary

Added columns in `street_segment_crash_attributes.csv` — OTIS Off-Peak Roadway Safety · Philadelphia 2020–2024

1. Hourly Crash Counts

Column	Type	Description
<code>h00 – h23</code>	integer	Number of crashes at each hour of the day (0 = midnight, 12 = noon). One column per hour; values are totals across all street segments.
<code>time_unknown</code>	integer	Number of crashes with no recorded time (<code>HOUR_OF_DAY_CLEAN</code> is NA, sourced from <code>TIME_OF_DAY</code> / <code>DIS</code>).

2. KSI Counts by Time Period

KSI = Killed or Suspected Serious Injury (`MAX_SEVERITY_LEVEL` 1 or 2, i.e. `is_ksi == 1`). All values are 2020–2024 totals.

Column	Type	Description
<code>ksi_peak</code>	integer	KSI crashes during peak hours (AM Peak 7–10 + PM Peak 16–19).
<code>ksi_offpk</code>	integer	KSI crashes during all off-peak hours (any non-peak timed crash).
<code>ksi_am</code>	integer	KSI crashes during AM peak (07:00–09:59).
<code>ksi_pm</code>	integer	KSI crashes during PM peak (16:00–18:59).
<code>ksi_midday</code>	integer	KSI crashes during midday (10:00–15:59).
<code>ksi_eve</code>	integer	KSI crashes during evening (19:00–21:59).
<code>ksi_night</code>	integer	KSI crashes during night/late (22:00–06:59).
<code>ksi_wknd</code>	integer	KSI crashes on weekends (Saturday–Sunday).
<code>ksi_wkdy</code>	integer	KSI crashes on weekdays (Monday–Friday).

3. Derived Rates

Rates expressed as percentages (0–100). NA when the denominator is zero — intentionally not coerced to 0 to avoid misleading values on segments with no crashes in a given period.

Column	Type	Formula	Description
<code>night_share</code>	numeric	$\text{night} / \text{total_crashes} \times 100$	Share of all crashes occurring during night/late hours (22–7).
<code>ksi_rate_peak</code>	numeric	$\text{ksi_peak} / \text{peak_crash} \times 100$	KSI rate among peak-hour crashes. NA if <code>peak_crash</code> = 0.
<code>ksi_rate_offpk</code>	numeric	$\text{ksi_offpk} / \text{offpk_crash} \times 100$	KSI rate among off-peak crashes. NA if <code>offpk_crash</code> = 0.
<code>ksi_rate_night</code>	numeric	$\text{ksi_night} / \text{night} \times 100$	KSI rate among night/late crashes. NA if <code>night</code> = 0.
<code>ksi_rate_wknd</code>	numeric	$\text{ksi_wknd} / \text{weekend} \times 100$	KSI rate among weekend crashes. NA if <code>weekend</code> = 0.

Notes

- Unit of analysis: road segment (`seg_id`). All counts are 2020–2024 totals. Divide by 5 for annual averages, consistent with existing `crash_py` / `ksi_py` columns.

- Time periods mirror feature engineering in Crash_Data_Analysis_Clean.qmd: AM Peak = 7–10, PM Peak = 16–19, Midday = 10–16, Evening = 19–22, Night/Late = 22–7.
- Only crashes with match_confidence == 'High' and a valid seg_id are included in segment aggregations.
- 75.9% of crashes have a valid hour (HOUR_OF_DAY_CLEAN); the remaining 24.1% contribute to time_unknown and are excluded from all hourly and period-based counts.